



DocuMind: Your Documents, Made Intelligent

A self-learning RAG system that turns any document collection into a conversational, context-aware knowledge base. Built with LlamaIndex, ChromaDB, Gemini, and LM Studio.

The Problem with Document Search



Keyword Search Fails

Traditional keyword search misses context and meaning, returning results based on exact matches rather than semantic understanding.



LLMs Hallucinate

Large language models generate confident answers without grounding in your documents, often fabricating details.



Knowledge Stale

Static document collections become outdated as the world changes—nothing updates automatically.

There had to be a smarter way.

What is DocuMind?

An intelligent Q&A system that retrieves from your documents, verifies answers for accuracy, and learns over time through continuous feedback and new research integration.



Retrieve

Find the most relevant context from any document collection using hierarchical chunking and semantic search.



Verify

Cross-validate answers using Gemini for accuracy and detect outdated information in your documents.



Learn

Ingest new research papers and apply corrections automatically to improve accuracy over time.

How It Works: The Full Pipeline

Layer 1: Ingestion

PDF and text documents parsed hierarchically into chunks (2048→512→128 tokens) and indexed in ChromaDB using HNSW for fast vector search.

Layer 2: Query Processing

Semantic cache checks for similar queries, auto-merging retriever combines related chunks, neural reranker filters to top 4 passages before LLM processing.

Layer 3: Enhancement

Chat history management, Gemini fact verification, Arxiv paper fetching, and analytics dashboard for performance insights.

Hierarchical Chunking & Auto-Merging Retrieval



Three-Tier Document Structure

- Level 1: 2048 tokens for major sections
- Level 2: 512 tokens for paragraphs
- Level 3: 128 tokens for sentences

Auto-merging retriever dynamically combines related chunks while neural reranker (MS-MARCO-MiniLM) scores and filters top 4 passages before the LLM ever sees them.

Self-Learning: Gemini Verification + Arxiv Integration

Gemini Fact Verification

- Every 5 queries, last 5 Q&A pairs sent to Gemini 2.5 Flash
- Detects conflicts and outdated information
- Optionally applies corrections directly to vector database

Arxiv Integration

- Key terms extracted from queries matched to cs.OS, cs.DC, and cs.PF categories
- Relevant papers fetched automatically with abstracts displayed
- User can add papers to knowledge base with one keystroke

Persistent Memory Across Sessions

1

Save to Disk

All sessions preserved—resume any conversation after a restart

2

Smart Summarization

Last 10 messages kept in full context; older messages auto-summarized into rolling summary (max 500 tokens)

3

Full Context

Follow-up questions understand complete conversation history

4

Commands

`/sessions` to list, `/history` to view, `/export` to save as JSON

Built for Speed: Semantic Caching & Analytics

Semantic Query Cache

92%

Similarity Threshold

Queries embedded with all-mpnet-base-v2

0

LLM Calls Saved

Instant cached response, no LLM call needed

Gemini responses also cached separately to reduce API costs significantly.

Analytics Dashboard

Access via **/analytics** command:

- Query count and cache hit rate tracking
- Average response rating metrics
- Source document usage frequency
- 1-5 star feedback collection per response

Two Modes: Privacy vs. Intelligence



Local Mode

- Fully offline—no data leaves your machine
- All processing on local GPU via CUDA
- Semantic cache and chat history functional
- Ideal for sensitive documents



Self-Learning Mode

- Gemini fact verification every 5 queries
- Automatic Arxiv paper recommendations
- Dynamic knowledge base updates
- Continuous accuracy improvement

Switch instantly with the `/mode` command

Tech Stack & What's Next



LlamaIndex



LM Studio (Qwen3-4B)



MS-MARCO-MiniLM



Arxiv API

Key Takeaways

- Local-first design means zero privacy trade-offs for offline use
- Self-learning loop makes the system more accurate with every session
- Modular architecture—swap any component independently



ChromaDB



all-mpnet-base-v2



Gemini 2.5 Flash



CUDA 12

What's Next

- Web UI / REST API wrapper
- Multi-user session support
- Finer domain-specific Arxiv category filtering